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Employing micro-aeration in anaerobic digestion of poultry litter and wheat straw: Batch kinetics and continuous performance

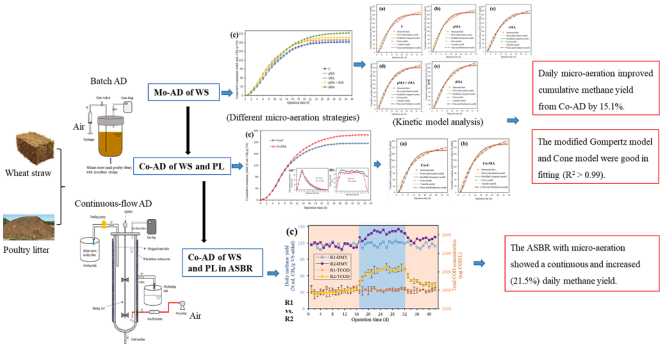
Yuanhang Zhan [a,*], Jun Zhu [a], Yiting Xiao [a], Leland C. Schrader [a], Sarah Xiao Wu [b], Ndeddy Aka Robinson Jr [b], Zhiwu Wang [c]

[a] Department of Biological and Agricultural Engineering, University of Arkansas, Fayetteville, AR 72701, USA
[b] Department of Chemical and Biological Engineering, University of Idaho, Moscow, ID 83844, USA
[c] Department of Biological Systems Engineering, Virginia Tech, Blacksburg, VA 24060, USA

HIGHLIGHTS

- Different micro-aeration strategies for anaerobic digestion (AD) were compared.
- Micro-aeration improved co-digestion of what straw and poultry litter.
- Methane yield kinetics of AD under micro-aeration were identified.
- Micro-aeration shortened the lag phase of AD.
- The continuous co-digestion process under micro-aeration was performed.

GRAPHICAL ABSTRACT



ARTICLE INFO

Keywords:
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ABSTRACT

In this study, different micro-aeration (MA) strategies for anaerobic digestion (AD) of poultry litter (PL) and wheat straw (WS) were examined. MA at different stages (pretreatment, middle, pretreatment plus middle, and daily) in batch AD of WS showed that daily MA had the highest increase (16.5 %) of the cumulative methane yield (CMY) compared to the control. Batch co-digestion (Co-AD) of WS and PL with daily MA obtained a furtherly improved (15.1 %) CMY of 225.44 N mL CH_4/g vS $_{added}$. The modified Gompertz model and Cone model were good in fitting the methane yield kinetics of MA engaged AD process (R^2 greater than 0.99). Daily MA shortened the lag phase of Co-AD by 3.4 %. The sequencing batch reactor for the Co-AD of WS and PL showed an increased (21.5 %) daily methane yield when 0.5-h/d MA was employed. The results provided support for the application of micro-aeration in the AD of agricultural wastes.

1. Introduction

Although anaerobic digestion (AD) has been widely applied to remediating agricultural wastes for producing methane containing biogas as a renewable energy resource, as reviewed by previous workers (Chen et al., 2008; Karki et al., 2021), the common AD process has

* Corresponding author.
E-mail address: yz062@uark.edu (Y. Zhan).

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several drawbacks such as low hydrolysis rate, unstable at high organic loading rate (OLR), and release of toxic/corrosive hydrogen sulfide. However, it has been reported recently that all these problems could be overcome through applying micro-aeration to the AD process as previously reviewed by Nguyen and Khanal (2018). Micro-aeration was defined as an advanced technology by dosing small quantities of air or oxygen into the AD reactor to enhance the process efficiency (Botheju et al., 2010). The air or oxygen might be dosed either one time by injection, intermittently (pulse-mode), or continuously, and the air or oxygen dosing might be conducted at different stages of the AD process (pretreatment, during digestion, or post digestion) (Girotto et al., 2018). Many researchers reported that with a small amount of energy input, micro-aeration showed great promise for improving AD process performance in organic degradation and methane production (Lim & Wang, 2013; D. Nguyen et al., 2019a; Xu et al., 2021). Zhen et al. (2021) applied micro-aeration pretreatment to the thermophilic (55 degrees Celsius (°C)) anaerobic digestion of corn stalk in a batch reactor (0.5 L) and found that micro-aeration promoted the generation of volatile fatty acids (VFA), and the digesters with a micro-aeration rate of 1 mL O_2/g volatile solids (VS) increased cumulative methane production by 13.2 %. While Fu et al. (2015) conducted oxygen injection to a batch AD (37 ℃) of corn straw after experiencing an obvious decrease in methane yield and observed an increase in the yield by 28.5 % and an improvement in the volatile solids (VS) removal efficiency by 29.4 %. Moreover, Fu et al. (2016) conducted another study on batch thermophilic AD of corn straw with daily micro-aeration and reported that the group with air injection rate of 12.5 mL/L_R per day improved the methane yield (16.5 %) and the vS removal efficiency. However, no direct comparisons between these micro-aeration strategies in terms of improving methane yield have been made. In addition, the substrate used for micro-aeration assisted AD process was mostly single substrate including primary sludge, wastewater, food waste or straw. There was a lack of research on the application of micro-aeration for anaerobic co-digestion (Co-AD) of agricultural waste, neither in batch nor in continuous flow operation mode. Thus, the effect of micro-aeration on methane yield efficiency and reaction kinetics of batch AD of agricultural wastes needs to be investigated. Also, the application of micro-aeration in continuous-flow AD of agricultural wastes is worth research as well.

In this study, the batch mono-digestion (Mo-AD) of wheat straw (WS) with different micro-aeration strategies were first examined and compared to select the best strategy with the highest methane yield. Then, the batch Co-AD of WS and poultry litter (PL) was conducted to furtherly improve the methane yield efficiency. Kinetics analysis of both Mo-AD and Co-AD was performed based on the batch experiments to mechanically analyze the effects of micro-aeration on methane yield performance. Lastly, the performance of the sequencing batch reactor for the continuous Co-AD of WS and PL with micro-aeration was also examined. Based on the results, this study aimed to support for the application of micro-aeration for enhancing methane production efficiency from AD of agricultural wastes.

2. Materials and methods

2.1. Substrate and inoculum sludge

2.1.1. Wheat straw and poultry litter

The substrate used in this study for both anaerobic mono-digestion (Mo-AD) and co-digestion (Co-AD) experiments was wheat straw (WS) collected from the University of Arkansas grass farm (Fayetteville, USA). The raw straw was then stored in a 4 degreesC walk-in refrigerator prior to pretreatment if not used immediately. Poultry litter (PL) was used as a nitrogen source to be co-digested with straw in the Co-AD process. The raw poultry litter used in this study was collected from the bedding waste in a chicken house in a poultry farm (University of Arkansas - Applied Broiler Research Farm, AR, US) and was then stored in a 4 ℃ walk-in refrigerator prior to pretreatment if not used immediately. The collected raw WS and PL was first mechanically cut using a mill (Food Processor, Hamilton Beach, Inc. VA, USA) before being screened using a 0.85 mm sieve (20 mesh) as suggested by Zahan et al. (2018b). The properties of the prepared WS and PL were summarized in Table 1.

2.1.2. Inoculum

The inoculum used for both the Mo-AD and the Co-AD was collected from a running anaerobic digester for methane fermentation and sieved through a 1.40 mm mesh to remove any large particles, after which it was then pre-incubated for seven days at 37 ℃ to remove any remaining biodegradable fraction of the sludge. The properties of the inoculum sludge were also shown in Table 1.

2.2. Micro-aeration

Different micro-aeration strategies were applied to the Mo-AD, including pretreatment micro-aeration (pMA, micro-aeration was employed in the initial stage); secondary micro-aeration (sMA, micro-aeration was employed in the middle stage at day 19 and day 20); pretreatment micro-aeration plus secondary micro-aeration (pMA + sMA); and daily micro-aeration (dMA, micro-aeration was employed daily). Five identical batch reactors were used for the Mo-AD of WS with different micro-aeration strategies employed, i.e., the control group (C, no micro-aeration employed), pMA, sMA, pMA + sMA, and dMA. The specific micro-aeration date and micro-aeration rate of different micro-aeration strategies were adopted from suggestions made by previous workers (Fu et al., 2016; Fu et al., 2015; Lim & Wang, 2013) and detailed in Table 2. The corresponding volume of air was injected into the batch reactors using a syringe. In the Co-AD experiments, the same daily micro-aeration was employed as in the Mo-AD.

2.3. Batch anaerobic digestion

2.3.1. Anaerobic mono-digestion of wheat straw

Five identical glass containers were used for Mo-AD of WS with different micro-aeration strategies employed. The glass container had a total volume of 648 mL, equipped with a gas outlet for biogas collection in Tedlar® gas bags (Tedlar Bag, CEL Scientific Corp., Cerritos, CA, USA). The glass container also had a tube reaching the sludge at the bottom of the reactor for nitrogen gas flushing and micro-aeration (air injection) as shown in Fig. 1 (a). To begin, 9.00 g of the wheat straw and 200 mL of the inoculum sludge were placed into the reactor to reach a ratio of $TS_{inoculum}/TS_{substrate}$ of 0.5 (Lim & Wang, 2013). Tap water was then added to the reactor to obtain a total working volume of 500 mL with pH adjusted to 8.00 with 1 N hydrochloric acid (HCl) solution and 1 N sodium hydroxide (NaOH) solution. Then, the reactor was purged with nitrogen gas for 5 min to eliminate oxygen before the reactor was fully sealed. After that, the Mo-AD batch reactors were operated with the corresponding micro-aeration strategies applied. The Tedlar® gas bags with biogas collected were replaced daily. The volume of the collected biogas was measured, and the biogas samples were analyzed for nitrogen gas (N_2), methane (CH_4), carbon dioxide (CO_2), etc. Pretrial experiments showed that the differences in the methane production performance between three reactors under the same circumstance were

Table 1
Physio-chemical properties of the substrate and the inoculum sludge.

Properties	What straw	Poultry litter	Inoculum sludge
TS	(96.68 ± 0.22) %	(90.70 ± 0.04) %	(2.18 ± 0.01) %
TVS	(92.64 ± 0.24) %	(71.99 ± 0.26) %	(16.47 ± 0.45) g/L
TC	(45.02 ± 3.25) %	(25.37 ± 1.33) %	–
TN	(0.76 ± 0.03) %	(3.37 ± 0.65) %	(205.7 ± 23.7) mg N/L
NH_4^+-N	–	–	(100.3 ± 5.0) mg N/L

Note: TS was total solids level (%); TVS was total volatile solids level (% substrate or g/L); TC, TN was total carbon content (%) and total nitrogen content (%), respectively; NH_4^+-N was ammonium nitrogen concentration (mg N/L).

Table 2
Operational parameters of the different micro-aeration strategies.

Experiment group	Micro-aeration operation Strategy	Micro-aeration date	Micro-aeration rate (N mL air/day)
C	Control	no	0
pMA	Pretreatment	Day 1–4	120
sMA	Secondary	Day 19–20	120
pMA + sMA	Pretreatment & secondary	Day 1–4 & day 19–20	120
dMA	Daily	Day 1–35	12

very small (<1%) (see Supplementary materials). Thus, the influence of systemic error could be ignored in this study.

2.3.2. Anaerobic co-digestion of wheat straw and poultry litter

Two identical glass containers with the same configuration as described above were used for Co-AD of WS and PL. For each reactor, the equivalent prepared WS and PL, 4.60 g each, were mixed and used as the substrate with 200 mL of the inoculum sludge added to reach the same ratio of $TS_{inoculum}/TS_{substrate}$ of 0.5 as in the Mo-AD experiments. The two batch reactors were operated as the control (Co-C) and the Co-AD with micro-aeration (Co-MA). The daily micro-aeration scheme was employed in the Co-AD experiments based on the results of the Mo-AD experiments. The operational protocol of the Co-AD experiments was the same as that of Mo-AD experiments as described above in Section 2.3.1.

2.3.3. Kinetic analysis

To examine the effect of micro-aeration on the kinetic parameters of AD process, five kinetic models: the first-order model; the modified Gompertz model; the Cone model, the transfer model; and the Chen and Hashimoto model were used to fit the methane yield from both the Mo-AD and Co-AD batch experiments in this study. The first-order kinetic model (Eq. (1)) was first described by Llabrés-Luengo and Mata-Alvarez (1987), who assumed that hydrolysis was the rate-limiting step that governed the overall process (Li et al., 2015; Zhang et al., 2019). However, the first-order model could not predict the conditions for maximum methane yield or lag phase (Kafle & Chen, 2016). The modified Gompertz model (Eq. (2)) was first used to predict methane production rate in a batch digester corresponding to the specific growth rate of methanogenic bacteria (Lay et al., 1998), and later became a widely used empirical regression model that described cell density in terms of exponential growth rates and lag phase duration, which could be used to identify the maximum methane production potential and lag phase (Kafle & Chen, 2016). Besides, recent research has reported that the Cone model (Eq. (3)), which is an empiric model that can estimate the methane yield rate and the maximum cumulative methane yield, is better than the other models in fitting the performance data in methane production from co-digestion (El-Mashad, 2013; Zahan et al., 2018b). Some recent studies also used the Transfer model (Eq. (4)) and Chen and Hashimoto model (Eq. (5)) to fit the methane production data from anaerobic co-digestion process (Karki et al., 2022).

$$P(t) = Y_m * \left(1 - e^{-kt}\right) \tag{1}$$

$$P(t) = Y_m * exp\left\{ - exp\left[\frac{R_m * e}{Y_m}(\lambda - t) + 1\right]\right\} \tag{2}$$

$$P(t) = \frac{Y_m}{1 + (-kt)^{-n}} \tag{3}$$

$$P(t) = Y_m * \left\{1 - exp\left[\frac{-R_m}{Y_m}(t - \lambda)\right]\right\} \tag{4}$$

$$P(t) = Y_m * \left(1 - \frac{k_{CH}}{HRT * \mu_m + k_{CH} - 1}\right) \tag{5}$$

where P (t) = the cumulative methane yield (mL CH_4/g VS) at the AD process operation time t (day); Y_m = the maximum methane yield potential (mL CH_4/g VS); e = 2.718; k = the hydrolytic rate constant (day^{-1}); t = the digestion time (day); R_m = the maximum methane production rate (mL CH_4/g VS/day); λ = the lag phase (day); n = the shape constant, which reveals whether there is a lag phase in the reactor; k_{CH}: Chen and Hashimoto constant (dimensionless); μ_m: Maximum specific growth rate of microorganisms (1/day); HRT: Digestion time or hydraulic retention time (days).

In this study, the correlation coefficients (R^2), Root Mean Square Prediction Error (rMSPE), and Akaike's Information Criterion (AIC), and

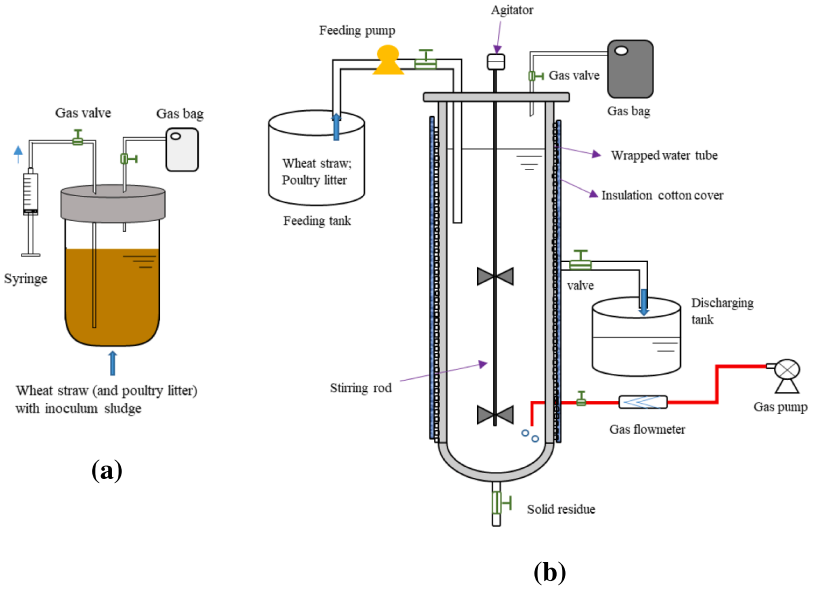


Fig. 1. The schematic of (a) the batch reactor and (b) the anaerobic sequencing batch reactor (ASBR) for digestion of wheat straw (and poultry litter) (under micro-aeration).

the Bayesian information criterion (BIC) were used to compare the three models under investigation. R^2 is known as a fitness index of the models and can be calculated using Eq. (6) (Shen & Zhu, 2017).

$$R^2 = 1 - \frac{SS_{res}}{SS_{tot}} = 1 - \frac{\sum_{i=1}^{m}(P_i - M_i)^2}{\sum_{i=1}^{m}(M_i - \overline{M})^2} \quad (6)$$

where R^2 = the correlation coefficient; SS_{res} = the sums of squares of residuals; SS_{tot} = the total sum of squares of deviations; P_i = the predicted value of point i by the model; M_i = the measured value of point i; $\overline{M}$ = the mean of the measured value; m = the number of measurements.

The rMSPE is a widely used indicator of the standard deviation between the predicted and measured values (Zhang et al., 2019). The model with a lower rMSPE value demonstrates a better fitting. The calculation of rMSPE was described in the following equation (Wang et al., 2011):

$$\text{rMSPE} = \sqrt{\sum_{i=1}^{m}\frac{(P_i - M_i)^2}{m}} \quad (7)$$

where rMSPE = Root Mean Square Prediction Error; P_i = the model predicted value of point i; M_i = the measured value of point i; m = the number of measurements.

AIC and BIC are alternatively used to compare models based on information technology development. The model with lower values of AIC and BIC is more likely to be suitable for fitting the experimental data (Motulsky & Christopoulos, 2004; D. D. Nguyen et al., 2019a). The formula for calculating AIC and BIC were given below.

$$\text{AIC} = \text{m*ln}\left(\frac{SS_{res}}{m}\right) + 2*(N+1) + \frac{2*(N+1)*(N+2)}{(m-N-2)} \quad (8)$$

$$\text{BIC} = \text{m*ln}\left(\frac{SS_{res}}{m}\right) + N*\ln(m) \quad (9)$$

where m = the number of measurements; SS_{res} = the sums of squares of residuals; N = the number of model parameters. In this study, N was 2, 3, and 3 for the first-order kinetic model, the modified Gompertz model, and the Cone model, respectively.

The difference (%) between the predicted value of maximum methane yield potential (mL CH_4/g VS) by a model and the measured value of the final cumulative methane yield (mL CH_4/g VS) was also used to evaluate the accuracy of the model prediction, the value of which could be calculated using Eq. (10).

$$\text{Difference}\,(\%) = \frac{Y_m - CMY_m}{CMY_m} * 100\% \quad (10)$$

where Y_m = the maximum methane yield potential (mL CH_4/g VS) predicted by the model; CMY_m = the cumulative methane yield measured (mL CH_4/g VS).

2.4. Continuous co-digestion in anaerobic sequencing batch reactor

Two identical cylindrical anaerobic sequencing batch reactors (ASBR) (Fig. 1 (b)) were used for the Co-AD of PL and WS in a continuous mode. The total volume of each reactor was 18 L with a full working volume of 16 L and 2 L headspace for maintaining enough biogas partial pressure. Detailed design of the ASBR could be found in a previously published article (Zhan et al., 2022), in which the ASBR was started up by gradually increasing OLR, and then the ground WS and PL with a weight ratio of 1:1 were loaded to the ASBR with water to reach a substrate total solids level (TS, %) of 4 %. The ASBR was operated following a hydraulic retention time (HRT) of 16 days. Based on the properties of PL and WS summarized in Table 1, the daily loading weight of PL and WS was found to be 16.72 g for both. The exchange volume of the liquid digestate to the loaded substrate was 0.75 L daily to satisfy the operation parameters above. The organic loading rate (OLR) was calculated to be 1.72 g VS/ (L_R·d) in this study (L_R was the reactor working volume in L).

The two identical ASBRs were operated simultaneously using the same substrate loading parameter and monitored for three periods. The first period (period 1) was the initial stage before micro-aeration was applied, which lasted for 16 days (1 HRT). In the subsequent period of 16 days, daily micro-aeration was applied, and in the third period of 10 days, micro-aeration was not employed. For comparison, the first reactor (R1) was the blank in which no micro-aeration was applied. While daily micro-aeration was applied to the second reactor (R2) in the second period (period 2) by injecting air using an air pump (Uniclife Air Pump UL40, Uniclife @ Amazon.com) for 0.5 h/day at an air flow rate of 0.75 L per hour controlled by a flow meter, which was 13.46 N mL air/g vS In each period, intermittent mixing was conducted twice daily, each for 10 min, and the solid sludge level after each period was controlled at 4 L. The produced biogas was collected in Tedlar® gas bags (Tedlar Bag, CEL Scientific Corp., Cerritos, CA, USA) through the top outlet (Fig. 1), and the gas bag was replaced daily. The collected biogas for each day was measured in volume and composition. In the meantime, the daily discharged effluent from the discharging outlet (Fig. 1) was sampled (125 mL) for physio-chemical analysis of total chemical oxygen demand (TCOD, mg/L). The effluent in the last five days of each period when the digestion process had entered a steady state was also analyzed for other physio-chemical parameters including soluble chemical oxygen demand (sCOD, mg/L), total volatile fatty acids (TVFA, mg HAc/L), and total alkalinity (ALK, g $CaCO_3$/L).

2.5. Sample analysis

2.5.1. Physio-chemical parameters analysis

Total solids (TS) and total volatile solids (TVS) were analyzed following the gravity method (Apha, 1998) using an oven (BINDER Inc., Bohemia, NY, USA) setting at 105 ℃ and a muffle furnace (Cole-Parmer, Vernon Hills, IL, USA) setting at 550 ℃, respectively. Total chemical oxygen demand (TCOD), soluble chemical oxygen demand (sCOD), total volatile fatty acids (TVFA), total alkalinity (ALK) of the liquid samples were analyzed following the standard methods (Apha, 1998). All these parameters were measured in triplicates and the mean values with standard deviations were obtained.

2.5.2. Biogas analysis

The biogas volume was measured using a wet gas test meter (Model XMF-1, Shanghai Cixi Instrument Co., ltd, China). The biogas volume was normalized using a gas factor of 0.92 calculated by (Olojede et al., 2018). The collected gas in gas bags was sampled and the composition was analyzed using a Shimadzu Gas Chromatograph (GC 2014, Shimadzu Scientific Instruments, Inc., Maryland, CO, USA) equipped with a thermal conductivity detector (TCD) and a ShinCarbon column (Restek, length 2 m × ID 2 mm). The operation of GC was described by Shen and Zhu (2017). The percentage of each gas composition could be obtained from the percentage value of peak area, with a peak corresponding to a specific gas.

2.6. Data analysis

The data in this study were recorded, processed, and graphed using Excel 2016 (Microsoft, WA, USA). The kinetic model was fitted using the solver tool in Excel 2016. Statistical significance was determined by analysis of variance (ANOVA) using JMP software (JMP Pro Version 12.0, SAS Institute Inc., NC, USA) with a p-value of 0.05. Tukey's Honest Significant Difference (HSD) test was used to compare differences of the performance of the two ASBRs in different periods.

3. Results and discussion

3.1. Effect of different micro-aeration strategies on mono-digestion of straw

The volumetric variation of daily methane production (DMP, N mL), the methane content (MC, %) in the produced biogas, and the cumulated methane yield (CMY, N mL CH_4/g vS $_{added}$) during batch operation of the anaerobic mono-digestion (Mo-AD) of straw are summarized and shown in Fig. 2 (a), (b), and (c), respectively. As shown in Fig. 2 (a), pMA, pMA + sMA, and dMA all improved the maximum DMP with reduced lag times. Compared with the control (C), which had the maximum DMP (123.64 N mL) on day 8, pMA and pMA + sMA arrived at the maximum DMP (143.23, and 135.24 N mL, respectively) on day 6, while dMA reduced the time to reach the maximum DMP (125.20 N mL) on day 7. The results indicated that micro-aeration at the initial stage (pMA or dMA) of straw digestion could improve the methane production efficiency by increasing the maximum methane yield as well as reducing the time to reach the maximums. In addition, both sMA and pMA + sMA showed obvious increase of DMP (94.8 % and 67.0 %, respectively) on day 20 (Fig. 2 (a)) after the secondary micro-aeration, indicating that micro-aeration during the middle stage of Mo-AD could further improve the methane production efficiency. Data of dMA showed a higher DMP than that of C during the operation, implying that daily micro-aeration could continuously improve methane production, which was consistent with the conclusion of Fu et al. (2016) that batch AD of corn straw with daily micro-aeration had better daily methane yield than the batch AD without micro-aeration.

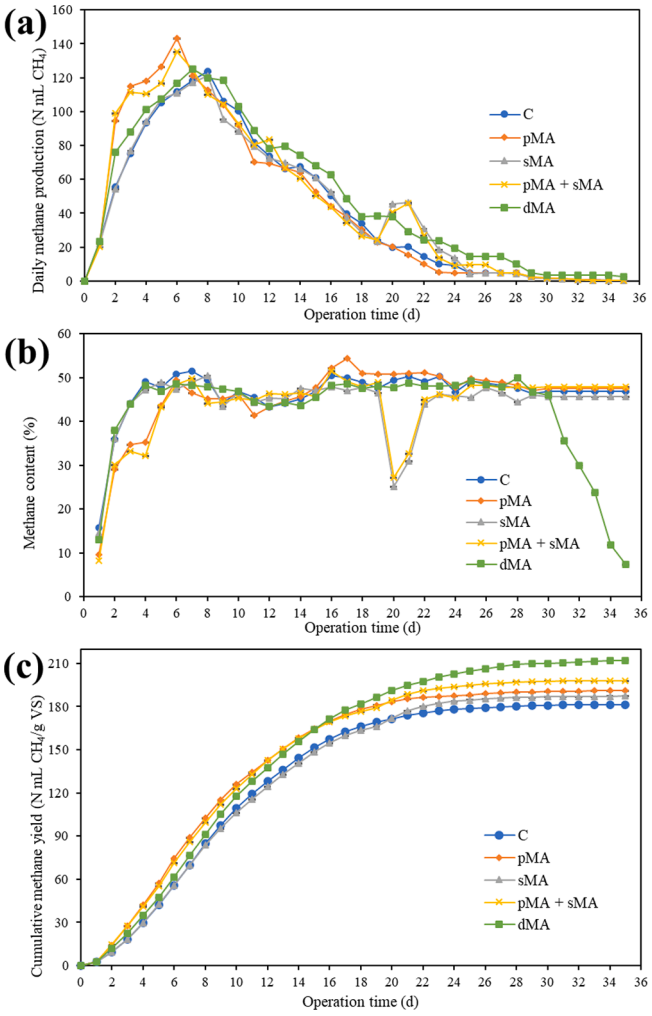


Fig. 2. (a) Daily methane production, N mL CH_4; (b) Methane content in biogas, %; (c) Cumulative methane yield, N mL CH_4/g vS $_{added}$, in batch reactors of anaerobic mono-digestion of wheat straw with different micro-aeration strategies employed. Note: C: the control, no micro-aeration; pMA: pretreatment micro-aeration; sMA: secondary micro-aeration; dMA: daily micro-aeration.

Micro-aeration could also affect the methane content (MC) in the produced biogas. As shown in Fig. 2 (b), the MC in pMA and pMA + sMA was obviously lower than that in C in the first 5 days when micro-aeration was applied. In addition, sMA and pMA + sMA also showed lower MC than that of C on day 20 and 21 when micro-aeration was applied during the middle stage (from day 19 to day 20). However, the MC was soon recovered to a similar level of C after micro-aeration was stopped. The results implied that micro-aeration in either the pretreatment stage or the middle stage could temporarily decrease the MC in the biogas. The observation that the MC of dMA dropped sharply after day 30 was largely caused by the decrease of methane production activity in combination of the injection of air to the digester, which contained a large percentage of nitrogen gas that could not be used, leading to a rapid decrease of MC in the collected biogas.

Lastly, as shown in Fig. 2 (c), the CMY were 181.44, 191.01, 187.24, 200.25, and 211.29 N mL CH_4/g vS $_{added}$ for C, pMA, sMA, pMA + sMA, and dMA, respectively. All the groups with micro-aeration had relatively higher CMY than C, indicating that all the micro-aeration strategies enhanced the CMY of Mo-AD of WS. This was consistent with the findings from previous studies that reported enhanced methane yield of AD process with micro-aeration applied in different stages (Amin et al., 2021; Fu et al., 2016; Fu et al., 2015). All the micro-aeration strategies improved methane production performance. In this study, the highest CMY was obtained in dMA, with an increase of 16.5 % compared with that of C. The results pointed out that, among the four micro-aeration strategies investigated in this study, daily micro-aeration was the most efficient one in improving methane yield in Mo-AD of WS. However, the observed CMY of Mo-AD with dMA was still relatively low as opposed to the findings in many other studies (Li et al., 2015). Nonetheless, improvement in methane yield could be achieved by Co-AD process of WS with PL based on the results from previous research (Li et al., 2020; Paranhos et al., 2020). Therefore, Co-AD of WS with PL under micro-aeration was further investigated.

3.2. Effect of micro-aeration on co-digestion of wheat straw with poultry litter

The variations of DMP (N mL CH_4), MC (%), and the CMY (N mL CH_4/g vS $_{added}$) of anaerobic co-digestion (Co-AD) of WS with PL using two groups of batch reactors were shown in Fig. 3 (a), (b), and (c), respectively, including the control group (Co-C), and the group with daily micro-aeration employed (Co-MA). First, the Co-AD process improved the methane production efficiency from the Mo-AD. The CMY of Co-C (195.84 N mL CH_4/g vS $_{added}$) was 7.9 % higher than that of C (181.44 N mL CH_4/g vS $_{added}$). The Co-AD group with micro-aeration (Co-MA) also showed an increased CMY (6.7 %) as opposed to the Mo-AD group with the same micro-aeration strategy (dMA). The results indicated that Co-AD of WS with PL had a better methane production efficiency than Mo-AD of WS. This observation was consistent with the results from many previous studies that methane production could be enhanced by co-digestion of agricultural wastes with animal waste (Karki et al., 2021; Paranhos et al., 2020).

Besides, similar to the findings in the Mo-AD process, micro-aeration also improved methane production for the Co-AD process. Compared to Co-C, the Co-MA had a slightly higher maximum DMP (158.01 N mL vs 151.83 N mL) but a shorter digestion time (6 days vs 7 days) to reach the maximum. Furthermore, the CMY of Co-MA (225.44 N mL CH_4/g vS $_{added}$) was improved by 15.1 % (see Supplementary materialsl), indicating an enhanced methane production efficiency of Co-AD by daily micro-aeration, as it was seen in Mo-AD. As a result, the Co-AD of WS

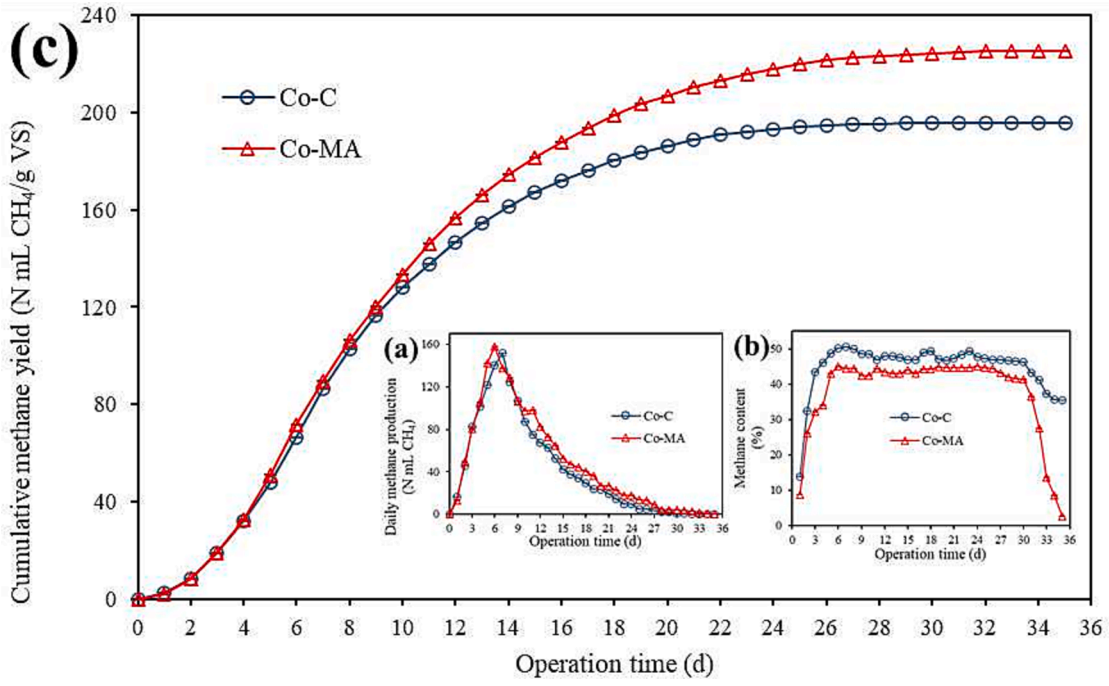


Fig. 3. (a) Daily methane production, N mL CH_4; (b) Methane content in biogas, %; (c) Cumulative methane yield, N mL CH_4/g vS_{added} in batch reactors for the anaerobic co-digestion of wheat straw and poultry litter, including the control (Co-C) and the group with daily micro-aeration (Co-MA).

with PL assisted with daily micro-aeration showed the highest methane production efficiency among all the groups investigated in this study.

3.3. Kinetic analysis

3.3.1. Model evaluation

Five models including the first-order kinetic model, the modified Gompertz model, the Cone model, the transfer model and the Chen and Hashimoto model were used to fit the measured data of batch MO-AD and Co-AD experiments. The calculated parameters for evaluation of the fitted models are summarized in Table 3. In general, all five models showed good fitting with $R^2 > 0.9590$. However, the modified Gompertz model and the Cone model had better goodness-of-fit than the other three models according to the comparisons of model evaluation parameters as shown in Table 3 and the comparisons of model fitting curves (see Supplementary materials). Previous studies also identified that the first-order kinetic model (Xu et al., 2021; Zhen et al., 2021) or the modified Gompertz model (Amin et al., 2021; Zhen et al., 2021) could be used to simulate the methane production kinetics of AD process under micro-aeration.

The R^2 values of the first-order kinetic model (ranging from 0.9590 to 0.9774), the Chen and Hashimoto model (ranging from 0.9722 to 0.9842) were obviously lower than those of the transfer model (ranging from 0.9856 to 0.9939) and were even lower than those of the modified Gompertz model (ranging from 0.9979 to 0.9997) and the Cone model (ranging from 0.9979 to 0.9996). The Cone model had slightly higher R^2 values (0.9994, 0.9996) than the modified Gompertz model (0.9980, 0.9987) in the Co-AD experiments, indicating that the Cone model fitted slightly better in methane yield of the Co-AD process. This observation was consistent with the reports from previous studies (El-Mashad, 2013; Li et al., 2015). Also, the modified Gompertz model and the Cone model had lower rMSPE values (ranging from 1.03 to 3.17 and from 1.48 to 2.64, respectively) than the first-order kinetic model (ranging from 9.16 to 13.06), the Chen and Hashimoto model (ranging from 7.64 to 10.25), and the transfer model (ranging from 4.75 to 6.95), suggesting a good correlation between the predicted and the measured values by the two models. The lower values of AIC and BIC of the modified Gompertz model (from 11.66 to 89.99, and from 13.00 to 91.33, respectively) and the Cone model (from 36.56 to 77.28, and from 37.89 to 78.61, respectively) demonstrated that they were more suitable for fitting the experimental data than the first-order kinetic model (both larger than 160), the Chen and Hashimoto model (both larger than 150), and the transfer model (both larger than 120). Finally, the overall Difference (%) was the smallest for the modified Gompertz model (-0.13 to 1.08), followed by the Cone model (4.39 to 9.37), the transfer model (4.53 to 10.02), the first-order kinetic model (10.23 to 19.33) and the Chen and Hashimoto model (35.57 to 51.46). The variation curves of the five models (see Supplementary materials) again illustrated that the modified Gompertz and Cone models correlated with the measured data much better than the first-order kinetic model, the transfer model, and the Chen and Hashimoto model. Therefore, these two models could be considered more suitable than the other three models in simulating the methane yield kinetics of micro-aeration assisted Mo-AD and Co-AD processes due to the relatively higher R^2 and lower rMSPE, AIC, BIC, and Difference (%). And the Cone model was slightly better for the Co-AD process than the modified Gompertz model.

3.3.2. Kinetic constants

The obtained kinetic constants from the three models such as the maximum methane yield potential (Y_m), maximum methane yield rate (R_m), hydrolysis rate constant (k), lag phase duration (λ), the shape factor (n), the Chen and Hashimoto constant (k_{CH}) and the maximum specific growth rate of microorganisms (μ_m) in different experiments were summarized, compared, and used to assess the effect of micro-aeration on the kinetic performance of AD processes (Table 3).

The maximum methane yield (Y_m) is a predictor of methane production potential based on substrate usage (Shamurad et al., 2020). It could be observed from Table 3 that the variation trend of Y_m for all models were consistent with the variation of measured CMY in both Mo-AD and Co-AD experiments, and Y_m was improved by different micro-aeration methods to varying degrees as discussed in Section 3.1 and 3.2, respectively. The predicted Y_m by the modified Gompertz model with a lower value of Difference (%) was closer to the measured CMY than those by the first-order kinetic model and the Cone model in all

Table 3
Kinetic parameters of the first-order kinetic model, modified Gompertz model, cone model, the transfer model and the Chen and Hashimoto model for the mono-digestion (Mo-AD) and co-digestion (Co-AD) process.

Group		Mo-AD					Co-AD	
		C	pMA	sMA	pMA + sMA	dMA	Co-C	Co-MA
Measured CMY (N mL/g VS_{added})		181.44	191.01	187.24	200.25	211.29	195.84	225.44
The first-order model	Y_m (mL/g VS_{added})	209.66	210.55	222.22	226.24	252.14	220.39	261.52
	k (d^{-1})	0.073	0.088	0.066	0.077	0.063	0.082	0.070
	R^2	0.9611	0.9709	0.9699	0.9774	0.9739	0.9590	0.9666
	rMSPE	11.39	9.82	10.33	9.16	10.75	12.48	13.06
	AIC	177.06	166.71	170.26	161.81	173.01	183.45	186.63
	BIC	177.39	167.05	170.60	162.15	173.34	183.79	186.97
	Diff (%)	15.56	10.23	18.69	12.98	19.33	12.53	16.01
The modified Gompertz model	Y_m (mL/g VS_{added})	182.42	191.36	189.25	201.05	212.88	195.58	224.84
	R_m (mL/g VS_{added}/day)	14.10	15.26	13.05	14.26	14.09	16.56	16.76
	λ (d)	2.11	1.36	1.85	1.23	1.74	2.06	1.99
	R^2	0.9997	0.9988	0.9986	0.9981	0.9992	0.9987	0.9980
	rMSPE	1.03	1.98	2.26	2.64	1.89	2.25	3.17
	AIC	11.66	57.17	66.43	77.31	54.02	66.14	89.99
	BIC	13.00	58.51	67.76	78.64	55.35	67.48	91.33
	Diff (%)	0.54	0.18	1.08	0.4	0.75	−0.13*	−0.27*
The cone model	Y_m (mL/g VS_{added})	192.05	202.68	203.47	217.35	231.08	204.43	239.24
	k (d^{-1})	0.115	0.129	0.106	0.116	0.100	0.124	0.112
	n	2.38	2.15	2.15	1.99	2.08	2.44	2.25
	R^2	0.9982	0.9979	0.9983	0.9986	0.9989	0.9994	0.9996
	rMSPE	2.44	2.64	2.49	2.3	2.21	1.54	1.48
	AIC	71.91	77.28	73.22	67.64	64.77	39.48	36.56
	BIC	73.24	78.61	74.55	68.97	66.10	40.82	37.89
	Diff (%)	5.85	6.11	8.67	8.54	9.37	4.39	6.12
The transfer model	Y_m (mL/g VS_{added})	195.64	200.89	205.35	213.87	232.46	207.39	242.13
	λ (day)	1.82	1.52	1.77	1.50	1.71	1.82	1.86
	R_m (mL/g VS_{added}/day)	19.96	23.46	18.77	21.81	20.26	23.70	24.01
	R^2	0.9856	0.9915	0.9899	0.9939	0.9915	0.9873	0.9907
	rMSPE	6.94	5.31	5.98	4.75	6.15	6.95	6.87
	AIC	144.97	126.23	134.53	118.46	136.48	145.06	121.04
	BIC	146.31	127.56	135.86	119.79	137.81	146.39	122.37
	Diff (%)	7.83	5.17	9.67	6.80	10.02	5.90	4.53
Chen and Hashimoto model	Y_m (mL/g VS_{added})	262.53	258.95	280.59	282.47	320.03	271.80	326.02
	k_{CH}	6.52	5.97	7.53	7.13	8.20	5.58	6.51
	μ_m (1/day)	0.57	0.66	0.57	0.65	0.59	0.57	0.55
	R^2	0.9722	0.9776	0.9803	0.9842	0.9832	0.9723	0.9803
	rMSPE	9.63	8.62	8.36	7.64	8.63	10.25	10.01
	AIC	167.87	160.11	157.94	151.68	160.21	172.26	170.58
	BIC	169.20	161.45	159.27	153.01	161.55	173.59	171.92
	Diff (%)	44.70	35.57	49.86	41.06	51.46	38.79	44.61

* The negative value means that the predicted value is lower than the measured value.

groups. For the Mo-AD experiments, Y_m was increased by all micro-aeration strategies and was the largest in dMA. For the Co-AD experiments, Y_m was higher in both Co-C and Co-MA than in Mo-AD groups (C and dMA, respectively), indicating that the co-digestion process improved the methane yield potential for the substrates used in this study. Besides, micro-aeration (Co-MA) achieved a higher Y_m compared to Co-C, demonstrating its ability to enhance the methane yield potential in the Co-AD process.

The hydrolysis rate constant (k), a parameter used to characterize the degradation rate of the substrate, can be calculated for the first-order kinetic model and the Cone model. A larger k value meant a higher degradation rate (Zhen et al., 2015). In general, the k value obtained from the Cone model (ranging from 0.102 to 0.142 d^{-1}) was slightly higher than that from the first-order model (from 0.063 to 0.100 d^{-1}). Considering the same variation of the k value between the two models and that the Cone model fitted better than the first-order kinetic model, different groups was compared with reference to the k value of the Cone model. For the Mo-AD group, pMA had the largest k value (0.129 d^{-1}), followed by pMA + sMA (0.116 d^{-1}), and they both were larger than the k value of C (0.115 d^{-1}). However, sMA and dMA both showed a lower k value (0.106 and 0.103 d^{-1}, respectively) than C. These results indicated that different micro-aeration strategies produced different impacts on the hydrolysis rate constant (k), i.e., pretreatment micro-aeration could enhance the hydrolysis rate, but secondary micro-aeration and daily micro-aeration had a negative effect on the k value. Despite that, they all improved the methane yield (Y_m) compared to C. The improvement of methane production due to pretreatment micro-aeration could be attributed to enhanced hydrolysis, but the reason for the improvement of methane production associated with the secondary micro-aeration was not straightforward, and could result from other pathways such as digestion of refractory organic matters, scavenging hydrogen sulfide by methanogens (Krayzelova et al., 2015) to maintain VFAs conversion (Fu et al., 2015). And daily micro-aeration could improve methane production by maintaining low VFAs concentration, or by promoting microbial activities (Fu et al., 2016) due to the small but continuous dosage of oxygen. For the Co-AD, the k value of Co-C (0.124 d^{-1}) and Co-MA (0.112 d^{-1}) were larger than that of C and dMA in the Mo-AD group, indicating that co-digestion improved substrate hydrolysis. Karki et al. (2021) concluded that co-digestion process could achieve enhanced digestibility due to synergistic effects of co-substrates and better process stability compared to the mono-digestion process. In addition, daily micro-aeration again showed a negative effect on k value, which was lower in Co-MA than in Co-C.

The constants of R_m, the maximum methane yield rate (N mL CH_4/g vS_{added} /d), and λ, the lag phase duration (d), both obtained from the modified Gompertz model or the transfer model, were important parameters to evaluate the AD process. A higher value of R_m and a shorter value of λ indicated a higher maximum methane production activity.

Since the modified Gompertz model had a better fitting than the transfer model as mentioned above, the value of R_m and λ obtained from the modified Gompertz model were used for comparisons between different groups. For the Mo-AD group, pMA and pMA + sMA had larger R_m values (15.26 and 14.26 N mL/g vS $_{added}$/d, respectively), while sMA and dMA had lower ones (13.05 and 14.06 N mL/g vS $_{added}$/d, respectively), compared to C (14.10 N mL/g vS $_{added}$/d), implying that pMA increased, sMA decreased, and dMA slightly decreased the maximum

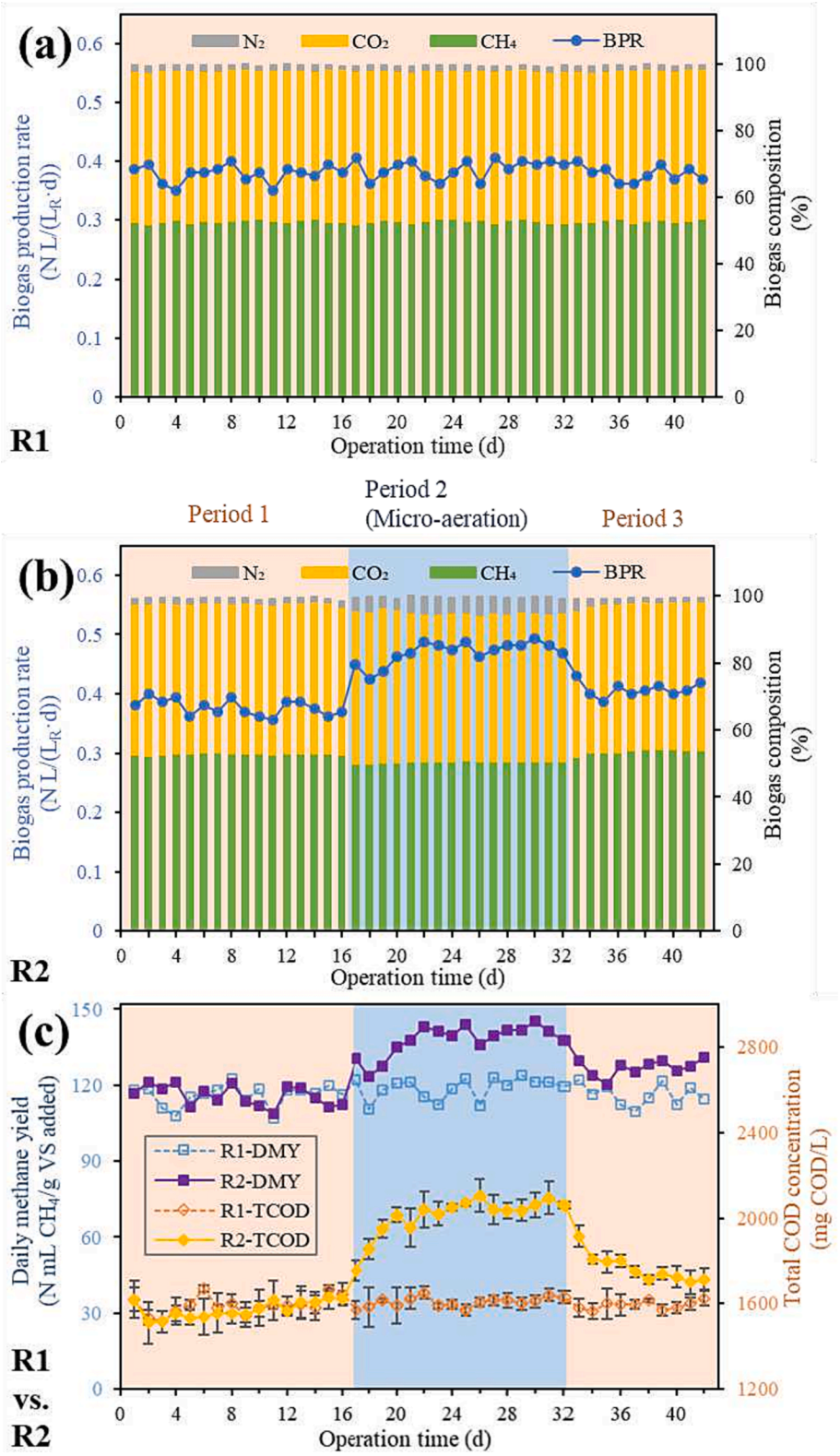


Fig. 4. Performance of the blank (R1) and treated (R2) anaerobic sequencing batch reactors for co-digestion of wheat straw and poultry litter in periods before, with, and after micro-aeration (period 1, 2, 3, respectively): Biogas production rate (BPR), N L/(L_R·d), and the biogas composition (%) in (a) R1 and (b) R2; (c) Comparison of daily methane yield (DMY), N mL CH_4/g VS_{added}, and the total chemical oxygen demand (TCOD, mg COD/L) in the effluent of R1 and R2.

methane production activity. This observation was analogous to the analysis of the difference between the k values mentioned above and could be explained by the different effects caused by the different micro-aeration strategies. In addition, all micro-aeration strategies reduced the lag phase time (λ, d), compared to C (2.11 d) by 1.23 d, 1.36 d, 1.74 d, and 1.85 d for pMA + sMA, pMA, dMA, and sMA, respectively. For the Co-AD group, the R_m values (16.56 to 21.90 N mL/g vS $_{added}$/d) were generally larger than those in the Mo-AD group, which again implied that Co-AD achieved better methane production activity than Mo-AD. In addition, Co-MA had a 3.4 % shorter λ (1.99 d) than Co-C (2.06 d), indicating that daily micro-aeration also improved methane production activities in the Co-AD process.

3.4. Performance of the continuous co-digestion of wheat straw and poultry litter with micro-aeration

Although the methane yield was generally higher in batch operation mode of AD reactors, the continuous-flow mode was normally employed in AD plants for undisrupted energy production (Zahan et al., 2018a). Thus, the performance of micro-aeration assisted Co-AD of WS and PL in continuous-flow mode was further investigated using ASBR based on the results of the batch experiments. The effect of daily micro-aeration on the ASBR performance was examined by monitoring and comparing the two reactors in different running periods. The daily variation of the biogas production rate (BPR, L/(L_R·d)) and the biogas composition (including contents (%) of methane (CH_4), carbon dioxide (CO_2), and nitrogen (N_2)) for R1 and R2 were shown in Fig. 4 (a) and (b), respectively. The comparison of the daily methane yield (DMY, N mL CH_4/g vS $_{added}$) and the total chemical oxygen demand (TCOD) variation between R1 and R2 were shown in Fig. 4 (c). And the performance of the two reactors in the steady state of each period were compared and summarized in Table 4.

For the blank reactor R1, the variation was basically stable in the three periods as shown in Fig. 4 (a) and (c). The average values of BPR, methane (CH_4) content, CH_4/CO_2 ratio (v/v) and DMY were (0.38 ± 0.01) N L/(L_R·d), (52.78 ± 0.51) %, 1.16 ± 0.02, and (117.23 ± 4.32) N mL CH_4/g vS $_{added}$, respectively. And the effluent analysis also showed relatively stable readings of TCOD (Fig. 4 (c)). The average content of TCOD, soluble COD (sCOD), total volatile fatty acids (TVFA), and total alkalinity (ALK) in the effluent were (1597.62 ± 31.61) mg /L, (727.78 ± 26.47) mg /L, (272.11 ± 6.45) mg HAc/L and (1.24 ± 0.04) g $CaCO_3$/L, respectively.

Table 4
The performance of the anaerobic sequencing batch reactor (ASBR) for continuous-flow co-digestion of wheat straw and poultry litter.

Parameters	R1 (Average)	R2 Period 1	Period 2 (Micro-aeration)	Period 3
Biogas production rate (N L/(L_R·d))	0.38 ± 0.01c	0.38 ± 0.01c	0.48 ± 0.01a	0.41 ± 0.01b
CH_4 content (%)	52.78 ± 0.51b	52.85 ± 0.27b	50.60 ± 0.11c	54.10 ± 0.17a
CH_4/CO_2 ratio (v/v)	1.16 ± 0.02b	1.17 ± 0.01b	1.13 ± 0.02c	1.22 ± 0.02a
Daily methane yield (N mL CH_4/g vS $_{added}$)	117.23 ± 4.32c	115.89 ± 4.02c	140.75 ± 2.82a	128.05 ± 2.20b
Effluent total COD (mg/L)	1597.62 ± 31.61c	1572.19 ± 39.50c	2047.08 ± 38.28a	1724.17 ± 18.28b
Effluent soluble COD (mg/L)	727.78 ± 26.47c	720.83 ± 37.04c	1216.67 ± 95.85a	913.33 ± 39.39b
Effluent total VFA (mg HAc /L)	272.11 ± 6.45c	265.20 ± 9.98d	430.10 ± 7.29a	306.30 ± 7.83b
Effluent total alkalinity (ALK) (g $CaCO_3$/L)	1.24 ± 0.04c	1.27 ± 0.04c	1.55 ± 0.03a	1.38 ± 0.05b
VFA/ALK ratio (g HAc equivalent to g $CaCO_3$ equivalent)	0.22 ± 0.01b	0.21 ± 0.01b	0.28 ± 0.01a	0.22 ± 0.01b

*Data are shown as mean ± standard error of five sampling days of the steady state in each period with three replicate measurements for each parameter (n = 15). Data in each row with different letters ("a" - highest value and "d" - lowest value) indicate significant differences ($p < 0.05$).

For R2, in general, the ASBR had a significantly higher BPR in period 2 when micro-aeration was applied but a slightly lower CH_4 content in the produced biogas (Fig. 4 (b)), resulting in a significantly higher DMY compared to that in period 1 and period 3 (Fig. 4 (c)). In period 1, the relatively stable performance of the continuous ASBR was similar to that of R1 (no significant difference, p greater than 0.05), showing a BPR of (0.38 ± 0.01) N L/(L_R·d), a CH_4 content of (52.85 ± 0.27) %, and a DMY of (115.89 ± 4.02) N mL CH_4/g vS $_{added}$. After micro-aeration was applied in period 2, the methane production performance was significantly enhanced ($p < 0.05$). The BPR ((0.48 ± 0.01) N L/(L_R·d)) was increased by 26.9 % and DMY ((140.75 ± 2.82) N mL CH_4/g vS $_{added}$ was increased by 21.5 %, while the CH_4 content slightly decreased to (50.60 ± 0.11) %. The methane production of R2 with micro-aeration in period 2 was also significantly higher than that of R1 in the same period. The DMY of R2 was significantly higher (18.34 %) than that of R1 in period 2 (Fig. 4 (c)). Moreover, when micro-aeration was stopped in period 3, the methane production efficiency showed a significant decline compared to in period 2. The BPR dropped to (0.41 ± 0.01) N L/(L_R·d) and the DMY decreased to (128.05 ± 2.20) N mL CH_4/g vS $_{added}$, but, interestingly, the CH_4 content was recovered to (54.10 ± 0.17) %. The results indicated that daily micro-aeration significantly enhanced methane production performance in the continuous-flow ASBR for co-digestion of WS and PL. The observation of enhanced methane production performance by micro-aeration was consistent with the findings from previous studies that applied micro-aeration to the continuous-flow AD process in other reactor configurations. For instance, Zhu et al. (2022) employed micro-aeration in a semi-continued AD reactor under different organic loading rates (OLRs) and reported that micro-aeration increased the median daily methane yields by 6 %, 10 %, and 8 % at OLRs of 1.1, 1.5, and 2.1 g VS/(L·day), respectively. D. Nguyen et al. (2019a) adopted intermittent (every 24 h), ORP-controlled micro-aeration for the AD of lignocellulosic biomass in a continuous-stirred tank reactor (CSTR) and found that a stable digester performance was obtained with an improved methane yield due to the rapid VFA conversion and methane production carried out by facultative anaerobes and hydrogenotrophic methanogens.

The CH_4 content in the produced biogas of R2 was slightly reduced by micro-aeration. The slightly lower CH_4 content in period 2 could be attributed to the nitrogen gas (N_2) from the injected air by micro-aeration, evidenced by the relatively higher N_2 content during period 2 than during period 1 or 3 (negligible) as observed in Fig. 4 (b) when the biogas composition was similar between R1 and R2 in period 1. In addition, the content of carbon dioxide (CO_2) was similar in period 1 and 2 (no significant difference), but relatively lower in period 3. As a result, the CH_4/CO_2 ratio was significantly lower in period 2 (1.14 ± 0.01) than in period 1 (1.17 ± 0.01) considering a slight decrease in CH_4 content. The lower CH_4/CO_2 ratio might be caused by the enhancement of methane production through hydrogenotrophic pathway, which converted CO_2 to CH_4 with hydrogen (H_2). D. Nguyen et al. (2019b), Zhen et al. (2021), and Wu et al. (2021) all reported that micro-aeration could facilitate hydrogenotrophic methanogenesis as the main pathway for methane production.

As for the effluent from R2, both the TCOD, sCOD, and total volatile fatty acids (TVFA) concentrations in the steady-state were significantly higher ($p < 0.05$) in period 2 than in period 1 and 3 and were increased by 30.2 %, 68.8 %, and 62.2 %, respectively, which indicated that micro-aeration could enhance hydrolysis and substrate solubilization during the continuous-flow co-digestion in the ASBR. In addition, the R2 effluent in period 2 also had a relatively higher total alkalinity (ALK) than in period 1 and period 3. The calculated TVFA/ALK ratio in R2 was

0.28 ± 0.01 in period 2, which was higher than in period 1 (0.21 ± 0.01) and period 3 (0.22 ± 0.01). Since the TVFA/ALK ratio was commonly used as an indicator for evaluating AD system stability (Tang et al., 2022) and the values of TVFA/ALK in different periods of R2 in this study were between 0.1 and 0.4, this indicated a favorable system condition without the risk of acidification (Zahedi et al., 2017). The effluent properties of R1 were similar to those of R2 in period 1(Fig. 4 and Table 4).

Besides, the methane production performance in R2 was higher in period 3 than that in period 1, though there was no micro-aeration in either period. R2 in period 3 showed significant increases ($p < 0.05$) in BPR, DMY, and CH_4 content by 7.9 %, 10.5 %, and 9.7 %, respectively, compared to that period 1. And the CH_4/CO_2 ratio reached a relatively higher value of 1.22 ± 0.01 in period 3. Moreover, the TCOD, sCOD, and TVFA contents in the R2 effluent in period 3 were also relatively higher than those in period 1(Table 4), indicating a higher substrate hydrolysis in period 3. The results implied that micro-aeration could improve the methane production performance of the AD process not only in the micro-aeration stage but also in the subsequent anaerobic stage of the ASBR. Similar results were reported by D. Nguyen et al. (2019b) that the improvement in methane yield due to intermittent micro-aeration (every 24 h) continued to be present in the subsequent anaerobic stage even after micro-aeration was stopped.

4. Conclusions

Daily micro-aeration could increase the CMY most in Mo-AD of WS by 16.5 %. The CMY was 15.1 % improved (225.44 N mL CH_4/g vS_{added}) from the Co-AD of WS and PL with micro-aeration. The modified Gompertz model and Cone model were able to accurately fit the methane yield kinetics data of the micro-aeration assisted AD process (R^2 greater than 0.99). With micro-aeration, the ASBR for continuous Co-AD of WS and PL achieved an enhanced performance with an increase in DMY by 21.5 %. The results indicated that micro-aeration in AD of agricultural wastes could be an appropriate technology for real applications.

CRediT authorship contribution statement

Yuanhang Zhan: Conceptualization, Methodology, Investigation, Writing – original draft. **Jun Zhu:** Validation, Supervision, Writing – review & editing. **Yiting Xiao:** Investigation, Resources. **Leland C. Schrader:** Methodology, Investigation. **Sarah Xiao Wu:** Resources. **Ndeddy Aka Robinson Jr:** Resources. **Zhiwu Wang:** Methodology.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.biortech.2022.128351.

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